

crmga — SaaS CRM Build Plan

Multi-tenant Laravel + Filament rebuild of the SuiteCRM 8.8.0 system · sold to other immigration firms

- 14-week v1
- 3 developers + PO
- Database-per-tenant
- Built with Claude Code
- Studio per tenant
- RESTful API + webhooks
- Per-tenant roles

Prepared 28 July 2026 · Example schedule below assumes kickoff **Mon 3 Aug** — replace Week 1 with your real start date.

1 · What we are building

A **multi-tenant SaaS CRM** (each customer company = its own isolated database) replacing the heavily-customised SuiteCRM. It is a **metadata-driven engine**: per-tenant metadata generates the schema, screens, permissions and API — which is what makes **Studio** (each company customising its own fields, layouts, relationships and modules, governed by you as super admin) possible. A modern **RESTful API** with OpenAPI docs and signed webhooks integrates WordPress, Meta/Instagram, WhatsApp, LinkedIn and any other stack. Each company defines its **own roles** (module × action × access level) with System Administrator and Regular User types. The CRM core is rebuilt on a clean schema (481 messy tables → ~35 tidy ones). First company live = Guinness & Associates.

2 · The team & who does what

Dev A
PLATFORM / METADATA & API LEAD

Architecture · database-per-tenant + provisioning · **metadata engine + SchemaManager (runtime DDL)** · super-admin panel + Studio governance · **RESTful API, OpenAPI, webhooks** · CI/CD · security · cutover.

Dev B
STUDIO & CRM UI

Studio UI (field/dropdown/layout/relationship/module builders) · **dynamic Filament rendering** from metadata · all CRM screens · role-matrix UI · DNC & Hot/Warm · dashboards · tenant settings.

Dev C
DATA, ACL & INTEGRATIONS

Core migrations · **ACL engine** (matrix, policies, query scopes) · data migration (ETL) + reconciliation · **WordPress + social integrations** · n8n rewrite · Asterisk/Vapi/SMS/email.

You
PRODUCT OWNER

Field-mapping answers · **Studio governance policy** · priorities · UAT sign-off · providing the sanitized dump (locally, Phase 7) + social/provider credentials (Phase 6).

3 · Timeline (14 weeks)

PHASE	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
P1 · Foundation + tenancy + metadata core		M1												
P2 · Core data model + ACL engine					M2									
P3 · STUDIO — per tenant								M3						
P4 · CRM UI + DNC + Hot/Warm + settings									M4					
P5 · RESTful API + webhooks										M5				
P6 · Integrations (social · WordPress · n8n)											M6			
P7 · Data migration / ETL (runs LOCALLY)												M7		
P8 · Hardening, UAT, go-live													M8	

◆ M1 metadata engine · M2 schema + ACL · M3 **Studio live** · M4 CRM screens · M5 REST API + webhooks · M6 integrations · M7 data migrated · M8 go-live. Internal beta after M4 (~wk 9).

4 · Phase detail — tasks, owners & "done when"

P1 Foundation + tenancy + metadata core

3 Aug – 14 Aug · Week 1–2 · Dev A (lead) · A · B · C

- [A] Scaffold Laravel 11 + Filament + stanc/tenancy + spatie + Passport/Sanctum + Horizon; central + tenant DBs; `tenant:create`; CI
- [A] **Metadata registry** (modules/fields/option-lists/layouts/relationships/change-log) + cached compiler
- [A] **SchemaManager** — safe runtime DDL: validate → plan → snapshot → apply → audit → cache bump (`_custom` sidecars)
- [A] Super-admin panel: create companies + **Studio governance** (mode, limits, change-request queue with DDL preview)
- [B] Tenant auth (+2FA) & **FieldTypeRegistry** (type → Filament component + cast + validation)
- [C] Load sanitized dump locally; profile data; refine field-map + enum lists

M1 Done when: tenant + field added via metadata → column in that tenant only, audited

P2 Core data model + ACL engine

17 Aug – 4 Sep · Week 3–5 · Dev C (lead) · A · B

- [A] Contactable base + `HasCustomFields`; polymorphic activities (Meeting/Note/Document/Email/Call/Task) + audit; `EmailAddress` morph
- [C] **ACL engine**: module × action × access level (All/Owner/Group/None), field-level ACL, Policies + shared query scopes, user types
- [C] Core entities: Company, Lead(vertical), Student, Assessment (CRS/FSW), StudyLead, LmiaCase, Client, Affiliate, Newsletter, SMS
- [B] Seed 29 starter roles per tenant; first end-to-end dynamic rendering from metadata

M2 Done when: Owner-level access enforced identically in UI and API; migrations clean

P3 STUDIO — per tenant

31 Aug – 25 Sep · Week 5–8 · Dev B (lead) · A

- [B] **Field Manager** (all types incl. relate) + **Dropdown Editor** (label convention preserved)
- [B] **Layout Editor** — drag-and-drop list/detail/edit/search, versioned layout JSON
- [B] **Dynamic rendering** hardening — `DynamicResource` builds form/table/infolist/filters; per-tenant metadata cache
- [A] **Relationship Manager** (1-M / M-M / 1-1) + **Module Builder** (tenant-defined modules w/ base, activities, permissions, API)
- [A] **Governance**: approval queue + DDL preview, audit, rollback, per-tenant limits, **blueprints** to push configs to new tenants

M3 Done when: a tenant customises itself with no deploy; super admin can gate/rollback

P4 CRM UI + DNC + Hot/Warm + settings

14 Sep – 2 Oct · Week 7–9 · Dev B (lead) · A · C

- [B] Screens for every entity on the dynamic renderer; activity timelines; global search
- [B] DNC filter/toggle; Hot/Warm toggles + dashboard widgets; daily count notifications
- [B] **Role-matrix UI** (SuiteCRM-like) + user management for tenant admins
- [A] Per-tenant settings (SMTP, telephony, branding, enabled modules) with encrypted secrets

M4 Done when: staff operate the CRM; tenant admin manages users/roles

P5 RESTful API + webhooks

28 Sep – 16 Oct · Week 9–11 · Dev A (lead) · C

- [A] **v1 REST** for every module incl. **Studio-created** — filtering/sort/includes/cursor pagination/idempotency
- [A] Auth & limits: OAuth2 + PATs + API keys with **scopes**; **OpenAPI 3.1 per tenant** + Swagger UI; rate limits + logs
- [A] **Outbound webhooks**: subscriptions, HMAC signing, retries/backoff, delivery log + replay
- [C] API contract tests + reference Postman/n8n collection

M5 Done when: external app CRUDs a Studio-made module + gets a signed webhook

P6 Integrations (social · WordPress · n8n)

12 Oct – 30 Oct · Week 11–13 · Dev C (lead) · A · B

- [C] WordPress plugin + form mappers (Gravity/CF7/WPForms/Elementor)
- [C] Meta Lead Ads + Instagram + WhatsApp Cloud, LinkedIn/TikTok/Google forms, generic ingest; shared FieldMapper + dedupe
- [C] Rewrite the 133 n8n workflows against the new REST API (pilot per family → waves)
- [A] Click-to-call (Reverb + per-tenant Asterisk AMI) + Vapi webhooks; SMS adapter; per-tenant SMTP/IMAP
- [B] In-CRM activity timeline (calls/SMS/emails) + integration settings UI

M6 Done when: Meta/WordPress lead → CRM → follow-ups → call tag-back

P7 Data migration / ETL (runs LOCALLY)

19 Oct – 30 Oct · Week 12–13 · Dev C (lead) · B

- [C] ETL sanitized dump → tenant DB: mapping, cleanup, UTC, primary_email join; idempotent + `--dry-run` + reconciliation
- [C] Import fields_meta_data + view defs → Studio metadata (existing customisation becomes tenant metadata)
- [B] Verify migrated data renders; fix field/display mismatches

M7 Done when: counts reconcile to the audit; existing custom fields appear in Studio

P8 Hardening, UAT, go-live

26 Oct – 6 Nov · Week 13–14 · Dev A (lead) · B · C

- [A] /security-review (tenant isolation, Studio DDL safety, API scopes); perf (Octane/FrankenPHP); backups + per-tenant export; cutover runbook
- [B] Finalise roles/permission matrix; UAT fixes; admin + Studio guide for tenant admins
- [C] Reconciliation re-run; full n8n cutover; monitoring/alerts

M8 Done when: UAT sign-off; security clean; first company LIVE

5 · Technology stack

PHP 8.3 & Laravel 11 · Filament v3 (staff UI) · stancl/tenancy (database-per-tenant) · spatie/permission (29 roles) · Laravel Passport (OAuth2 API) · MySQL/MariaDB · Redis + Horizon (jobs/scheduler) · Reverb (click-to-call) · FrankenPHP/Octane (speed) · Pest / PHPStan / Pint + GitHub Actions (tests & CI) · plus the custom metadata engine (SchemaManager · FieldTypeRegistry · DynamicResource), OpenAPI 3.1, and a signed webhook dispatcher.

6 · Decisions to lock BEFORE we start

BLOCKING — decide before Phase 1

- Timeline option — A: 3 devs / ~14 wks (this plan) · B: 4 devs / ~11–12 wks · C: staged (9 wks to go-live, Studio wks 10–15).
- Hosting / infra — where the SaaS runs + per-tenant backups (rec: Docker + FrankenPHP on a cloud VPS).
- Domain + wildcard DNS/TLS — *.crm.<domain> for companies + a central admin domain.
- Studio governance default — new tenants start in disabled / request-only / self-serve ? + limits (max fields/modules, may create relationships) (rec: request-only).
- Confirm Filament UI (no separate SPA) & keeping the source char(36) UUID primary keys (rec: yes to both).

SHOULD-DECIDE — by Phase 2–3

- Legacy adapter for the 133 n8n workflows — build a thin /Api/V8/* bridge (~3–4 days) as insurance and migrate in waves, or rewrite big-bang? (rec: build the bridge).
- Which social platforms are must-have in v1 — Meta/Instagram · WhatsApp · LinkedIn · TikTok · Google? (rec: Meta + WhatsApp + WordPress in v1, others via generic ingest).
- Built-in modules — any of Quotes/Invoices, Reports builder, Campaigns, Knowledge Base, Events, Projects in v1? (rec: no — tenants can build simple versions in Studio).
- Telephony model — per-company PBX/AMI credentials vs one shared (rec: per-company).
- Reporting v1 = Filament widgets + CSV/Excel export (rec: yes) · inbound email (IMAP) in v1 or later (rec: later).

PROVIDE LATER — per phase, not blocking now

- Social/WordPress API credentials (Meta app, WhatsApp, LinkedIn...) — Phase 6.
- SMS gateway choice (Twilio/other) + Vapi, Asterisk AMI, SMTP/IMAP credentials in `.env` — Phase 6.
- Sanitized data dump — used **locally** at Phase 7.
- Compliance / hosting region for immigration PII (PIPEDA/GDPR).
- Billing / subscriptions (Stripe) + self-serve signup — fast-follow after go-live.

Full rationale + recommendations are in `docs/ARCHITECTURE.md` §5. You can reply “go with the recommendations” or change specific items.